

Top 150 DSA Problems Roadmap (JavaScript)

This roadmap is organized by patterns to help you recognize problem-solving techniques quickly in coding interviews and LeetCode.

1. Arrays & Hashing (20)

Easy

1. Two Sum
2. Contains Duplicate
3. Valid Anagram
4. Intersection of Two Arrays
5. Concatenation of Array
6. Replace Elements with Greatest Element on Right Side
7. Is Subsequence
8. Length of Last Word

Medium

1. Group Anagrams
2. Top K Frequent Elements
3. Product of Array Except Self
4. Valid Sudoku
5. Encode and Decode Strings
6. Longest Consecutive Sequence
7. Sort Colors
8. Majority Element II

Hard

1. First Missing Positive
 2. Trapping Rain Water
 3. Candy
 4. Sliding Window Maximum
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2. Two Pointers (15)

Easy

1. Valid Palindrome
2. Merge Sorted Array
3. Remove Duplicates from Sorted Array
4. Squares of a Sorted Array

Medium

1. Two Sum II
2. 3Sum
3. Container With Most Water
4. Next Permutation
5. Sort Colors
6. Partition Labels
7. Rotate Array

Hard

1. Trapping Rain Water
 2. 4Sum
 3. Median of Two Sorted Arrays
 4. Minimum Window Substring
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3. Sliding Window (15)

Easy

1. Maximum Average Subarray I
2. Best Time to Buy and Sell Stock

Medium

1. Longest Substring Without Repeating Characters
2. Permutation in String
3. Find All Anagrams in a String
4. Minimum Size Subarray Sum
5. Longest Repeating Character Replacement
6. Fruit Into Baskets
7. Binary Subarrays With Sum
8. Subarray Product Less Than K

Hard

1. Minimum Window Substring
 2. Sliding Window Maximum
 3. Substring With Concatenation of All Words
 4. Count Vowel Substrings
 5. Shortest Subarray with Sum at Least K
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4. Stack (15)

Easy

1. Valid Parentheses
2. Implement Stack Using Queues
3. Baseball Game

Medium

1. Min Stack
2. Daily Temperatures
3. Evaluate Reverse Polish Notation
4. Next Greater Element II
5. Decode String
6. Asteroid Collision
7. Simplify Path
8. Car Fleet

Hard

1. Largest Rectangle in Histogram
 2. Basic Calculator
 3. Maximal Rectangle
 4. Remove Invalid Parentheses
-

5. Binary Search (15)

Easy

1. Binary Search
2. Search Insert Position
3. First Bad Version

Medium

1. Search in Rotated Sorted Array
2. Find Minimum in Rotated Sorted Array
3. Koko Eating Bananas
4. Time Based Key-Value Store
5. Capacity To Ship Packages Within D Days
6. Search a 2D Matrix
7. Find Peak Element

Hard

1. Median of Two Sorted Arrays
 2. Split Array Largest Sum
 3. Find in Mountain Array
 4. Count Smaller Numbers After Self
 5. Russian Doll Envelopes
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6. Linked List (15)

Easy

1. Reverse Linked List
2. Merge Two Sorted Lists
3. Linked List Cycle
4. Middle of the Linked List

Medium

1. Add Two Numbers
2. Remove Nth Node From End
3. Reorder List
4. Copy List With Random Pointer
5. Sort List
6. Swap Nodes in Pairs
7. Rotate List

Hard

1. Merge K Sorted Lists
 2. Reverse Nodes in K Group
 3. LFU Cache
 4. LRU Cache
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7. Trees (25)

DFS Basics

1. Maximum Depth of Binary Tree
2. Same Tree
3. Invert Binary Tree
4. Subtree of Another Tree
5. Balanced Binary Tree

BFS

1. Binary Tree Level Order Traversal
2. Right Side View
3. Average of Levels in Binary Tree

Medium

1. Validate Binary Search Tree
2. Lowest Common Ancestor
3. Kth Smallest Element in BST
4. Construct Binary Tree from Traversals
5. Path Sum II
6. Binary Tree Zigzag Level Order Traversal
7. Diameter of Binary Tree
8. House Robber III
9. Count Good Nodes in Binary Tree

Hard

1. Serialize and Deserialize Binary Tree
 2. Binary Tree Maximum Path Sum
 3. Recover Binary Search Tree
 4. Vertical Order Traversal
 5. Word Search II
 6. Sum of Distances in Tree
 7. Flatten Binary Tree to Linked List
 8. Redundant Connection II
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8. Heap / Priority Queue (10)

Medium

1. Kth Largest Element in Array
2. Top K Frequent Elements
3. Find K Closest Elements
4. Task Scheduler
5. Last Stone Weight
6. Reorganize String
7. Find Median From Data Stream

Hard

1. Merge K Sorted Lists
 2. Sliding Window Median
 3. IPO
-

9. Backtracking (10)

Medium

1. Subsets
2. Permutations
3. Combination Sum
4. Combination Sum II
5. Letter Combinations of Phone Number
6. Palindrome Partitioning
7. Generate Parentheses
8. Word Search

Hard

1. N-Queens
 2. Sudoku Solver
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10. Graphs (10)

BFS/DFS Graphs

1. Number of Islands

2. Clone Graph
3. Flood Fill
4. Pacific Atlantic Water Flow
5. Course Schedule
6. Rotting Oranges
7. Walls and Gates

Hard

1. Alien Dictionary
 2. Network Delay Time
 3. Word Ladder
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11. Dynamic Programming (20)

1D DP

1. Climbing Stairs
2. House Robber
3. House Robber II
4. Min Cost Climbing Stairs
5. Coin Change
6. Longest Increasing Subsequence
7. Partition Equal Subset Sum
8. Decode Ways

2D DP

1. Unique Paths
2. Longest Common Subsequence
3. Edit Distance
4. Target Sum
5. Distinct Subsequences
6. Interleaving String

Hard

1. Burst Balloons
 2. Regular Expression Matching
 3. Palindrome Partitioning II
 4. Maximum Product Subarray
 5. Word Break II
 6. Best Time to Buy and Sell Stock III
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Suggested Order

1. Arrays & Hashing
 2. Two Pointers
 3. Sliding Window
 4. Stack
 5. Binary Search
 6. Linked List
 7. Trees
 8. Heap / Priority Queue
 9. Backtracking
 10. Graphs
 11. Dynamic Programming
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Practice Strategy

First Pass

- Understand the pattern
- Solve easy and medium problems
- Skip hard if stuck

Second Pass

- Re-solve without hints
- Improve time complexity understanding

Third Pass

- Focus on speed and interview-style solving
 - Try solving mediums in under 25–30 minutes
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Important JavaScript Templates

Frequency Counter

```
const map = new Map()
```


DFS

```
function dfs(node) {  
  if (!node) return  
  
  dfs(node.left)  
  dfs(node.right)  
}
```

BFS

```
const queue = [root]
```

Sliding Window

```
let left = 0  
  
for (let right = 0; right < arr.length; right++) {  
  
}
```

Binary Search

```
while (left <= right) {  
  let mid = Math.floor((left + right) / 2)  
}
```

Final Goal

By the end of this roadmap you should be able to:

- Identify patterns quickly
- Solve medium-level LeetCode problems confidently
- Explain time complexity clearly
- Write JavaScript solutions comfortably in interviews